LI-DTI Web Tool Usage & Interpretation Guide

LI-DTI Web provides an interactive way to inspect individual LI-DTI predictions and the evidence supporting them.

LI-DTI predicts drug-target interactions by combining drug-based and target-based evidence. For a selected drug-target pair, the drug-based contribution is calculated from drugs related to the selected drug that are already known to interact with the selected target, while the target-based contribution is calculated from targets related to the selected target that are already known to interact with the selected drug. LI-DTI Web shows how these two sources of evidence contribute to the selected prediction.

These two representation models are built from drug-side and target-side similarity information. In the sunburst plot, these similarities appear in the outer ring and indicate which source of evidence supports each contribution.

Drug-side similarities:

- **chemical**: similarity between drug chemical structures.
- **ddi**: similarity derived from drug-drug interaction information.
- **drug-disease association**: similarity derived from drug-disease association profiles.
- **drug-side effect association**: similarity derived from drug-side effect association profiles.
- **drug-target interaction profile similarity / covariance**: similarity derived from known drug-target interaction profiles between drugs.

Target-side similarities:

- **sequence**: similarity between protein sequences.
- **ppi**: similarity derived from protein-protein interaction information.
- **protein-disease association**: similarity derived from protein-disease association profiles.
- **drug-target interaction profile similarity / covariance**: similarity derived from known drug-target interaction profiles between targets.

The sections below explain how to generate a sunburst plot and how to interpret each level of the visualization for a selected drug-target prediction.

How To Generate A Plot

1. Select a drug from the drug menu, or select a target from the target menu.
2. The second menu is updated with predicted partners ranked by LI-DTI score.
3. Select the drug-target pair of interest.
4. Review the prediction statistics and known interaction panels.
5. Click **Generate Explainability Plot**.

The plot appears below the button as a sunburst plot. Each sector represents part of the LI-DTI score for the selected drug-target pair.

Prediction Statistics

The web page reports three quantities before the sunburst plot.

Global Percentile compares the selected score with all LI-DTI scores. A higher percentile means that the selected pair receives a higher score than most drug-target pairs in the dataset.

Target Rank for Drug gives the rank of the selected target among all predicted targets for the selected drug.

Drug Rank for Target gives the rank of the selected drug among all predicted drugs for the selected target.

The known interaction panels list interactions already present in the dataset. These lists help explain which known drugs or known targets can contribute to the prediction.

How To Read The Sunburst Plot

The sunburst plot should be read from the center outwards.

Inner ring: drug and target representation models. The two central sectors show how much of the prediction score comes from the drug representation model and how much comes from the target representation model. A large drug sector means that the prediction is mainly supported by drugs similar to the selected drug. A large target sector means that the prediction is mainly supported by targets similar to the selected target.

Middle ring: similar drugs or similar targets. On the drug side, the middle ring shows drugs that are similar to the selected drug and are known to interact with the selected target. On the target side, the middle ring shows targets that are similar to the selected target and are known to interact with the selected drug.

Outer ring: type of evidence. The outer ring shows which similarity sources support each similar drug or target. Larger outer sectors indicate that a similarity source contributes more strongly to that local part of the prediction.

The percentages are local to their parent sector. For example, a percentage shown inside a similar drug sector describes that drug's share of the drug representation model. A percentage shown inside an evidence sector describes that evidence source's share of the corresponding similar drug or target.

Example Plots

Below are two examples that illustrate how to generate and interpret the sunburst plots. Both pairs can be selected from the **Quick Examples** section at the top of the page. After selecting a pair, click **Generate Explainability Plot** to display the sunburst plot.

Example 1: Hydroxychloroquine And Toll-like Receptor 8

This is a target-based example. To select it manually, first select **Toll-like receptor 8** in the target menu. The drug menu is then updated with predicted drugs for this target; select **Hydroxychloroquine** and click **Generate Explainability Plot**. The prediction statistics report **Global Percentile: 99.93th percentile**, **Target Rank for Drug: 1/422**, and **Drug Rank for Target: 1/548**.

In this example, LI-DTI predicts an interaction between Hydroxychloroquine and Toll-like receptor 8. The sunburst plot in Figure 1 shows that most of the score comes from the target representation model.

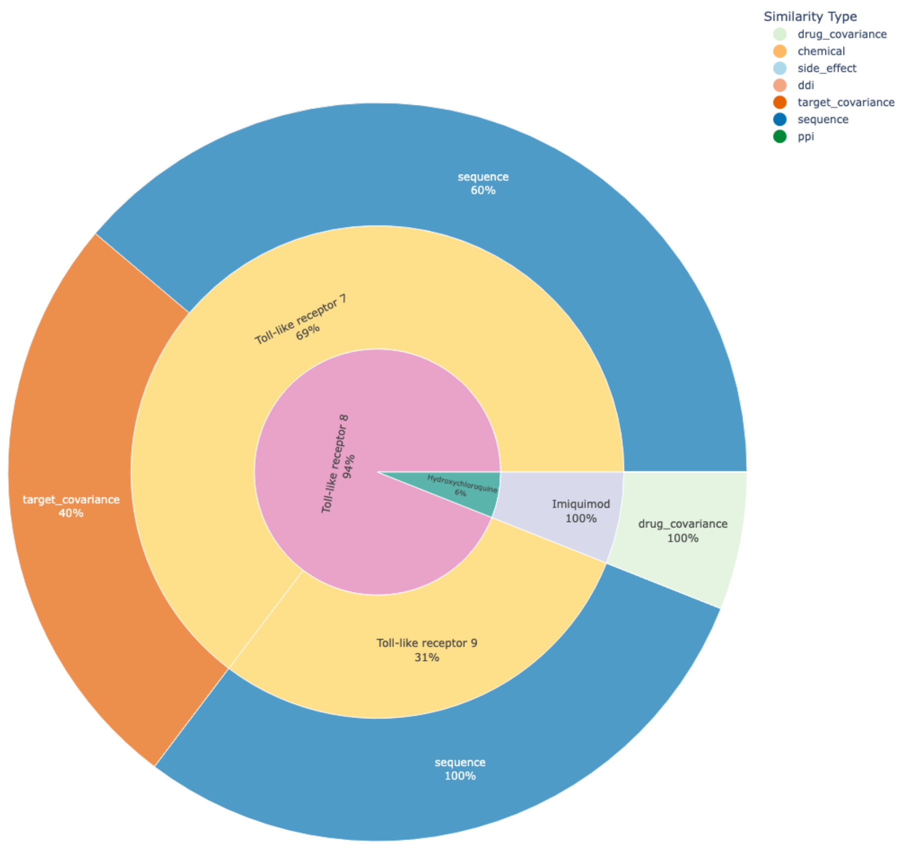

Figure 1: **Explanation of Hydroxychloroquine–Toll-like receptor 8 interaction predicted scores.** The sunburst plot provides the breakdown of the score for this DTI. The drug representation model accounts for 6% of the total score and the target representation model accounts for 94%. On the target side (Toll-like receptor 8), most of the score comes from similarities to Toll-like receptor 7 and Toll-like receptor 9. On the drug side (Hydroxychloroquine), the contribution comes from Imiquimod and is driven almost entirely by drug-target interaction profile similarity.

This means that the prediction is mainly explained by targets similar to Toll-like receptor 8 that are already connected to Hydroxychloroquine in the data. The largest target-side sectors are Toll-like receptor 7 and Toll-like receptor 9. The contribution from Toll-like receptor 7 is supported by sequence similarity and target covariance, while the contribution from Toll-like receptor 9 is supported by sequence similarity.

The drug representation model contributes only a small part of the score. This part comes from Imiquimod, a drug connected to Toll-like receptor 8 in the data. The outer ring shows that this drug-side contribution is driven almost entirely by drug-target interaction profile similarity.

Example 2: Trandolapril And Angiotensin-converting Enzyme 2

This is a drug-based example. To select it manually, first select **Trandolapril** in the drug menu. The target menu is then updated with predicted targets for this drug; select **Angiotensin-converting enzyme 2** and click **Generate Explainability Plot**. The prediction statistics report **Global Percentile: 99.61th percentile**, **Target Rank for Drug: 1/423**, and **Drug Rank for Target: 3/548**.

In this example, LI-DTI predicts an interaction between Trandolapril and Angiotensin-converting enzyme 2. The sunburst plot in Figure 2 shows that both representation models contribute substantially to the score: 59% from the drug representation model and 41% from the target representation model.

The drug-side contribution comes from Moexipril, a drug similar to Trandolapril that is known to interact with Angiotensin-converting enzyme 2. The outer ring shows that this contribution is supported by chemical similarity and drug-target interaction profile similarity.

The target-side contribution comes from Angiotensin-converting enzyme, a target connected to Trandolapril in the data and similar to Angiotensin-converting enzyme 2. The outer ring shows that this contribution is driven mainly by sequence similarity, with a smaller contribution from target covariance.

Practical Reading Checklist

When interpreting a new LI-DTI sunburst plot:

1. Start with the inner ring and identify whether the drug or target representation model contributes more.
2. Move to the middle ring and identify the largest similar drugs or similar targets.
3. Move to the outer ring and check which evidence sources support those large sectors.
4. Use the prediction statistics to understand where the selected pair ranks globally and within the selected drug or target.

The plot provides an explanation of the model score and is intended to support biological hypothesis generation. Experimental validation is required before treating any predicted interaction as confirmed.

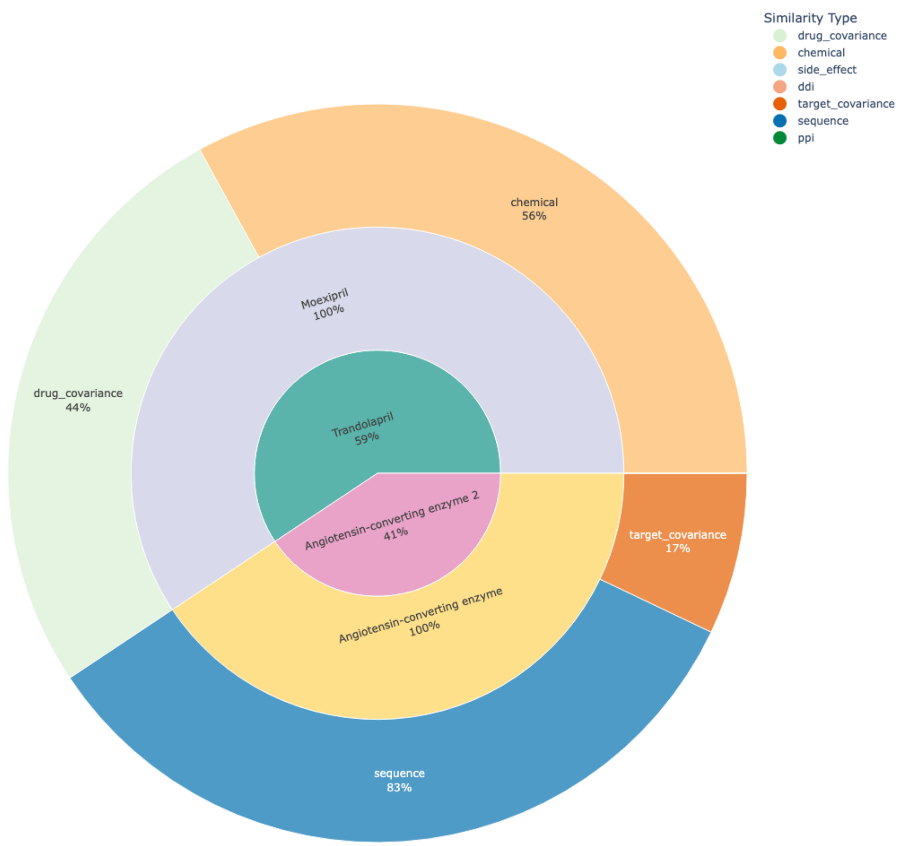

Figure 2: **Explanation of Trandolapril–Angiotensin-converting enzyme 2 interaction predicted scores.** The sunburst plot provides the breakdown of the score for this DTI. The drug representation model accounts for 59% of the total score and the target representation model accounts for 41%. On the drug side (Trandolapril), the contribution comes from Moexipril and is supported by chemical similarity and drug-target interaction profile similarity. On the target side (Angiotensin-converting enzyme 2), the contribution comes from Angiotensin-converting enzyme and is driven mainly by sequence similarity, with a smaller contribution from target covariance.